IIT Madras BS Degree & Internship Strategy

User: iit m bs dual diploma all courses

AI: IIT Madras offers online BS degree programs featuring multiple levels, including a Foundation level, dual **Diplomas** (Programming and Data Science), a BSc degree, and a full BS degree.

Program Levels and Structure
• **Foundation Level (32 Credits):** 8 mandatory courses covering basic Mathematics, Statistics, Computational Thinking, and Python programming.
• **Diploma Level (54 Credits total across two diplomas):**
- **Diploma in Programming:** 6 theory courses (5 core + 1 skill enhancement) plus 2 hands-on projects. Topics include Java, Database Management Systems (PostgreSQL), and modern application development.
- **Diploma in Data Science:** 6 theory courses (5 core + 1 skill enhancement) plus 2 hands-on projects. Topics include Machine Learning techniques, Business Analytics, and Data Science tools.
• **BSc Degree Level:** Requires earning 114 total cumulative credits.
• **BS Degree Level:** Requires earning 142 total cumulative credits by completing advanced project and elective courses.

User: am i able to take 4 courses and 2 project in my second term

AI: Yes, you are able to take 4 courses and 2 projects in a single term, provided you have already entered the Diploma Level and have a standard **Credit Clearing Capability (CCC) of 4**.

The academic guidelines cap your term registration at a **maximum of 4 theory courses**. Because the 2 course projects are considered separate components paired with specific diploma courses, they do not count against your 4-course theory limit.

Keep the following critical factors in mind:
• **Extreme Workload:** Managing 4 courses alongside 2 projects requires an estimated 70 to 90 hours of study per week.
• **Prerequisites:** You must ensure that you have cleared the specific prerequisite courses required before the term registration window opens.
• **Academic Standing:** If your CCC drops below 4 due to poor performance or pending backlogs, the portal will automatically restrict you from registering for the maximum load.

User: i want to do diploma in both give me a structured course selection in order

AI: To complete **both diplomas**, you must clear a total of **12 theory courses, 2 skill enhancement courses, and 4 projects**.

The Dual Diploma Master Schedule:
• **Term 1: Software & Logic Foundations**
- Theory 1: Database Management Systems (DBMS)
- Theory 2: Programming, Data Structures and Algorithms using Python (PDSA)
- Theory 3: Modern Application Development I (MAD 1)

- Theory 4: System Commands (SC — Programming Skill Enhancement)
- Project 1: MAD 1 Project
• **Term 2: Core Programming & Data Prep**
- Theory 1: Object-Oriented Programming Concepts using Java
- Theory 2: Modern Application Development II (MAD 2)
- Theory 3: Machine Learning Foundations (MLF)
- Theory 4: Business Data Management (BDM)
- Project 2: MAD 2 Project
- Project 3: BDM Project
- Milestone: Unlocks the Diploma in Programming.
• **Term 3: Machine Learning Core**
- Theory 1: Machine Learning Techniques (MLT)
- Theory 2: Machine Learning Practice (MLP)
- Theory 3: Tools in Data Science (TDS — Data Science Skill Enhancement)
- Theory 4: Business Analytics (BA)
- Project 4: MLP Project
• **Term 4: Advanced Elective & Wrap-Up**
- Theory 1: Deep Learning / Generative AI (or alternative elective)
- Milestone: Unlocks the Diploma in Data Science.

User: are two skill enchantment necessary

AI: Yes, both skill enhancement courses are absolutely mandatory if you want to earn both certificates.

Each individual diploma is strictly packaged with **5 core theory courses, 2 projects, and 1 specific skill enhancement course**.
• **For the Diploma in Programming:** You must take **System Commands (SC)**.
• **For the Diploma in Data Science:** You must take **Tools in Data Science (TDS)**.

If you skip a skill enhancement course, you will not be awarded that respective diploma certificate, and the portal will not allow you to advance to the BSc Degree Level via that pathway.

User: similarly give me term wise course for bsc and bs degree to complete them asap

AI: To fast-track your **BSc and BS Degrees** right after completing your Dual Diplomas, you must continuously maximize your **Credit Clearing Capability (CCC) of 4 theory courses per term**.

Phase 1: The BSc Degree Fast-Track (1–2 Terms)
• **Term 5 (BSc Level - Core Engineering)**
- Theory 1: Software Engineering
- Theory 2: Operating Systems & Computer Architecture
- Theory 3: Software Testing
- Theory 4: Big Data / Introduction to AI
• **Term 6 (BSc Wrap-Up & Transition to BS)**
- Theory 1: AI: Search Methods for Problem Solving
- Theory 2: Deep Learning
- Theory 3: BS Elective 1 (e.g., Natural Language Processing)

- Theory 4: BS Elective 2 (e.g., Computer Vision)
- Milestone: You graduate with your BSc Degree.

Phase 2: The BS Degree Sprint (1 Final Term)
• **Term 7 (BS Level - Specialization & Capstone)**
- Theory 1: Deep Learning in Practice
- Theory 2: BS Elective 3 (e.g., GenAI Foundations)
- Theory 3: BS Elective 4 (e.g., MLOps / Reinforcement Learning)
- Theory 4: BS Elective 5 (e.g., Design Thinking)
- Project: BS Capstone Project / Industry Apprenticeship
- Milestone: Full BS Degree in Data Science and Applications.

User: ok now if i want to get intern at chess.com by next summer idealy at the end of my second term of diploma should i do that by reducing the number of courses in in finaal 2 terms

AI: Yes, you should absolutely reduce your course load during your final two terms of the diploma if your primary goal is to land an internship at Chess.com by next summer.

• **The Timeline Mismatch:** Chess.com's Summer Internship applications and interviews happen between January and April, coinciding with your Term 2. Reducing your load allows time for applications.
• **Specialized Skillset Building:** Engineering internships focus heavily on practical development stacks (like Java/Go/React). Lowering course counts gives you time to make production-grade projects.
• **DSA and LeetCode Preparation:** You will need dedicated blocks of time to study Data Structures & Algorithms (DSA) outside of class.

User: if i have a solid project build for chess.com will the judge me based on my knowledge or completition of my diploma as technically i have not comple diploma 2 and just done with diploma 1

AI: They will judge you almost entirely on your knowledge, technical skills, and your project, rather than the exact completion status of your diploma.

• **'Enrolled Status' is Enough:** Chess.com requires applicants to be enrolled in higher education or a relevant tech program. Your active IIT Madras status covers this.
• **The Power of a Domain-Specific Project:** Building a project interacting with chess proves self-motivation and a strong product culture fit, separating you from standard resumes.
• **The Interview Reality:** The initial evaluation heavily checks technical foundations like SQL, Object-Oriented Logic, and system tools—all covered completely by your completion of Diploma 1 components.